Midterm 1

Graded

Student

Toby Hansen

Total Points

74.4 / 105 pts

Question 1

Q1: Translation

9.4 / 10 pts

– **0 pts** Correct: (a) and (b) are each worth one point. (b)-(F) are each worth two points.

✔ – **0.6 pts** major flaw (a)

– **0.6 pts** major flaw (b)

– **1.25 pts** major flaw (c)

– **1.25 pts** major flaw (d)

– **1.25 pts** major flaw (e)

– **1.25 pts** major flaw (f)

Question 2

Q2: Boolean Proof

10 / 10 pts

✔ **– 0 pts** Correct by truth table

– 0 pts Correct by propositional rules, including check

– 1 pt correct by rules, didn't verify that conclusion fits rules

– 1 pt rules, very small problem

– 2 pts rules, minor problems

– 2 pts truth table, minor problems

– 4 pts rules, somewhat convincing

– 4 pts truth table, major problems

– 4 pts rules, bad statement(s), forced answer to work

– 6 pts rules, far from a convincing argument

– 6 pts incomplete solution by truth table, bad reasoning

– 0.25 pts Wrote (possibly/fully) correct truth table but did not explain where the final conclusion comes from. Needs SOME explanation or at least circling of correct solution

– 1 pt Truth table settings of q_1, q_2, q_3, q_4 were not in standard order. If you use a different order you MUST explain WHY the 16 rows contain all of the possibilities

– 1 pt Truth table missing final most important column for $\wedge$. Just writing that 0010 is the only row for which I and II are both correct is not sufficient for a truth table proof. That's using an extra logical argument, which is not what a truth table proof does.

– 0.5 pts Missing the first 4 columns showing the settings of q_1, q_2, q_3, q_4
These are an important part of the truth table structure since they are what make it immediately easy to read off solutions.

Your table might contains the same information elsewhere but, not including that information on the Left Hand Side loses some of the usefulness of the formal truth table structure and it is therefore not a complete truth table.

– 8 pts fragmentary or almost entirely wrong

– 10 pts Missing solution or totally wrong.

Question 3

Q3: Predicate Dog Proof

20 / 20 pts

✔ **– 0 pts** Correct

– 14 pts Major errors

– 2 pts Small Error

– 10 pts Major errors (sometimes due to errors in translations)

– 5 pts Correct table and some correct ideas and but your proofs are missing too many important justifications

– 4 pts major problem

– 8 pts lots of problems, but some of the ideas

– 12 pts overall badly confused, but some progress

– 16 pts fragmentary or almost all wrong

– 20 pts no answer

Question 4

Q4a: Partial Orders

10 / 10 pts

✔ **+ 10 pts** Correct

+ 0 pts Missing or all wrong

+ 1 pt Correctly applied reflexivity to find $(a, a), (b, b), (c, c), (d, d) \in P$

+ 2 pts Correctly applied transitivity to find $(a, c) \in P$

+ 2 pts Correctly applied antisymmetry to find $(b, a), (c, b), (c, d), (c, a) \notin P$

+ 2 pts Correctly applied transitivity to find $(b, d) \notin P$

+ 1 pt Found $(b, d) \notin P$, but with insufficient / incorrect justification

+ 6 pts Mostly correct Part (i), but with extra entries

+ 6 pts Mostly correct Part (i), except for using $(a, c) \in P$ before proving it

+ 3 pts Correctly found all three possible P' and their Hasse diagrams

+ 2 pts Minor error in Part (ii), e.g., one missing / extra P's or one incorrect Hasse diagram

+ 1 pt Major errors in Part (ii)

+ 0 pts Missing or too many errors in Part (ii)

Question 5

Q4b: Equivalence Relations

9 / 10 pts

+ 10 pts Correct

+ 0 pts Missing or all wrong

✔ + 1 pt Correctly applied reflexivity to find $(a,a),(b,b),(c,c),(d,d),(e,e) \in Q$

✔ + 2 pts Correctly applied symmetry to find $(c,a),(c,b) \in Q$ and $(d,c),(e,c) \notin Q$

✔ + 2 pts Correctly applied transitivity to find $(a,b),(b,a) \in Q$

+ 1 pt Applied transitivity to find $(a,b) \in Q$ or $(b,a) \in Q$, but the proof is flawed, e.g., without applying symmetry to get $(c,a) \in Q$ or $(c,b) \in Q$ first

+ 3 pts Correctly applied transitivity to find $(a,d),(a,e),(b,d),(b,e) \notin Q$ and then symmetry for the 4 "flipped" pairs

✔ + 2 pts Applied transitivity to find pairs not in Q, but missing a few or incorrect / missing justification

1 They should be $\neg P(c,a)$ and $\neg P(c,b)$.

+ 7 pts Mostly correct Part (i), with a few missing / extra entries

+ 6 pts Correct table in Part (i), but missing too much justification

+ 5 pts Mostly correct table in Part (i), with a few missing / extra entries and missing too much justification

✔ + 2 pts Correctly found all two possible Q' and their resulting partitions

+ 1 pt Partially correct Part (ii), with either a few missing / extra Q' or incorrect / missing partitions (but not both)

+ 0 pts Missing Part (ii) or too many errors

Question 6

Q5a: Extended GCD Algorithm

4 / 10 pts

– **0 pts** Correct
a=13; b=--29

– **0 pts** wrote 29 instead of -29 on solution line

– **5 pts** Incorrect and missing important "shown work".
Please see solution key for more explanation

– **5 pts** Incorrect result and did not use method taught in class

✔ – **6 pts** Incorrect and/or missing "shown work".
Please see solution key for more explanation

– **2 pts** Correct final answer but did not use method taught in class. The problem instructions explicitly required the use of the method taught in class. In particular, we expected to see the intermediate results

$$10 = 96 - 2 \cdot 43$$
$$3 = -4(96) + 9 \cdot 43$$

written out separately.

Solutions that did not show these explicit intermediate results were not using the technique and formatting structure taught in class and had points deducted.

See the solution key for more details.

– **3 pts** Correct result but did not show understandable work or was missing some important steps in work

– **6 pts** Too much wrong/missing

– **7 pts** Too much wrong/missing

– **8 pts** Too much wrong/missing

– **10 pts** No solution or too much missing

Question 7

Q5b: Finding Inverses

0 / 5 pts

+ 5 pts Correct:
13 is an inverse of 96 modulo 43.
67 is an inverse of 43 modulo 96.

✔ **+ 0 pts** Missing solution or totally wrong

+ 2.5 pts Correct solution for inverse of 96

+ 2.5 pts Correct solution for inverse of 43

+ 2.5 pts The first answer is consistent with solution to part (a) (which was incorrect)

+ 2.5 pts The second answer is consistent with solution to part (a) (which was incorrect)

+ 1.5 pts Correct approach for the first inverse, with mistakes in the calculation

+ 1.5 pts Correct approach for the second inverse, with mistakes in the calculation

+ 3 pts Flipped the solutions

+ 1.5 pts Correct inverse of 96 but incorrect answer (not 13, which is smallest natural)

+ 1.5 pts Correct inverse of 43 but incorrect answer (not 67, which is smallest natural)

Question 8

Q5c: Chinese Remainder Theorem

0 / 5 pts

+ 5 pts Correct: $3 \cdot (-13) \cdot 23 + 4 \cdot 6 \cdot 50 = 303$

✔ **+ 0 pts** Missing or totally incorrect approach

+ 4 pts Correct approach, with mistakes in calculation

+ 4 pts Almost correct answer, but not the smallest natural

+ 4 pts Correct answer, without using the techniques taught in class

+ 3 pts Almost correct approach, but missing minus sign in -13.

+ 2 pts Almost correct approach, but switching the remainders 3 and 4 in the formula

Question 9

Q5d: EC. Solving the Quadratic

0 / 5 pts

+ 5 pts Correct: Found 4 solutions (1, 6, 29, and 34) and explained why they are all the solutions

✔ **+ 0 pts** Missing or totally incorrect solution

+ 4 pts Found all 4 solutions, but without sufficient justification for the completeness of solutions (i.e., no other solutions exist)

+ 3 pts Found 3 of the 4 solutions

+ 2 pts Found 2 of the 4 solutions

+ 1 pt Found 1 of the 4 solutions

+ 4 pts Found all 4 solutions, but also some incorrect ones

Question 10

Q6: True/False **12** / 20 pts

10.1 **(a)** **0** / 2 pts

– **0 pts** Correct

✔ – **2 pts** (a) wrong, should be FALSE

10.2 **(b)** **0** / 2 pts

– **0 pts** Correct

✔ – **2 pts** (b) wrong, should be TRUE

10.3 **(c)** **2** / 2 pts

✔ – **0 pts** Correct

– **2 pts** (c) wrong, should be FALSE

10.4 **(d)** **2** / 2 pts

✔ – **0 pts** Correct

– **2 pts** (d) wrong, should be TRUE

10.5 **(e)** **2** / 2 pts

✔ – **0 pts** Correct

– **2 pts** (e) wrong, should be FALSE

10.6 **(f)** **0** / 2 pts

– **0 pts** Correct

✔ – **2 pts** (f) wrong, should be FALSE

10.7 **(g)** **0** / 2 pts

– **0 pts** Correct

✔ – **2 pts** (g) wrong, should be FALSE

10.8 **(h)** **2** / 2 pts

✔ – **0 pts** Correct

– **2 pts** (h) wrong, should be TRUE

10.9 **(i)** **2** / 2 pts

✔ – **0 pts** Correct

– **2 pts** (i) wrong, should be TRUE

10.10 (j) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts (j) wrong, should be FALSE

NAME: Toby Hansen

SPIRE ID: 34834236

COMPSCI 250
Introduction to Computation
Midterm 1 – Fall 2026

M. Golin and M Qiao

15 October 2026

DIRECTIONS:

- Answer the problems on the exam pages.
- There are 6 problems on pages 2-11, some with multiple parts, for 100 total points plus 5 extra credit. Final scale will be determined after the exam.
- Page 12 contains useful definitions and is given to you separately – do not put answers on it!
- If you need extra space use the back of a page – both sides are scanned.
  But, if you do write on the back of a page, you must explicitly add a note on the front side stating that you are continuing on the back page. Otherwise, we might not see your solution on Gradescope.
- No books, notes, calculators, or collaboration.
- In case of a numerical answer, an arithmetic expression like "$2^{17} - 4$" need not be reduced to a single integer.
- Your answers must be LEGIBLE, and not cramped. Write only short paragraphs with space between paragraphs

1	/10
2	/10
3	/20
4	/20
5	/20+5
6	/20
Total	/100+5

Family Name: Hansen

Definitions for Questions 1-3: On an outing to the park one day, Blaze and Rhonda met their friends Clover, Lulu, and Phoebe, where all five dogs engaged in various play activities. The set of dogs $D = \{b, c, l, p, r\}$ included a subset of three golden retrievers $G = \{c, p, r\}$.

The four activities $A = \{DW, HSC, RB, WP\}$ consisted of Dog Wrestling, High Speed Chase, Retrieving a Ball, and Water Play.

The unary predicate $G(d)$ indicates that "dog d is a Golden", i.e., $d \in G$.

The binary relation $E \subseteq D \times A$ is defined by the predicate $E(d, x)$, meaning "dog d engaged in activity x".

Question 1 (10): (Translations)

- **(a, 1) Statement I (to English):** $\neg(E(l, RB) \rightarrow E(p, DW))$

  if Lulu engaged in Retriving a Ball, then Phoebe did NOT engage in Dog Wrestling

- **(b, 1) Statement II (to symbols):**

  If Lulu retrieved a ball, then Blaze did not engage in retrieving a ball and also Blaze did not engage in water play.

  $E(l, RB) \rightarrow (\neg E(b, RB) \wedge \neg E(b, WP))$

- **(c, 2) Statement III (to English):** $\forall d : G(d) \rightarrow (E(d, RB) \wedge E(d, WP))$

  All Golden retrievers engages in Retriving a ball, and engaged in water play

- **(d, 2) Statement IV (to symbols):**

  Rhonda engaged in every activity that Phoebe did, Phoebe engaged in every activity that Clover did, and Clover engaged in every activity that Lulu did.

  $\forall a : [(E(p, a) \rightarrow (E(r, a)) \wedge (E(c, a) \rightarrow E(p, a)) \wedge (E(l, a) \rightarrow E(c, a))]$

- **(e, 2) Statement V (to English):**

  $$\neg\exists d : \exists d' : (d \neq d') \wedge (\forall z : (E(d, z) \leftrightarrow E(d', z)))$$

  there are no two unique dogs that engaged in all of the exact same activities overall

- **(f, 2) Statement VI (to symbols):**

  Blaze engaged in exactly two of the activities.

  $\exists x : \exists y : (x \neq y) \wedge (\forall z : E(b, z) \leftrightarrow ((z = x) \vee (z = y)))$

Family Name: Hansen

Question 2 (10): (Boolean Proof) For this problem, please use the abbreviations $q_1 = E(b, RB)$, $q_2 = E(b, WP)$, $q_3 = E(l, RB)$ and $q_4 = E(p, DW)$.

Using *either* a truth table *or* a deductive sequence proof, **prove** that there is **exactly one** setting of the variables q_1, q_2, q_3, and q_4 satisfying **Statements I and II**.

Note that you must show both that your setting satisfies the two statements, and that no other setting does so. If you construct a complete correct truth table, you will prove both. If you use a deductive sequence so that your setting is implied by the two statements, you must *also* verify that your setting in fact does satisfy both the statements.

So that you may have correct inputs to Question 3, we will tell you that the correct setting has q_3 is **true** and the other three propositions are **false**.

q_1	q_2	q_3	q_4	$(\neg(q_3 \to q_4))$	$\wedge$	$(q_3 \to (\neg q_1 \wedge \neg q_2))$
0	0	0	0	0	0	1
0	0	0	1	0	0	1
0	0	1	0	1	1	1
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	0	1
0	1	1	0	1	0	1
0	1	1	1	0	1	1
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	1	0	1
1	0	1	1	0	0	1
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	1	0	1
1	1	1	1	0	0	1

There is only a single set of values for which statement I and statement II are both true.

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Question 3 (20) (Predicate Proof) : For this problem, we want you to use Statements I, II, III, IV, V, and VI to determine exactly which dogs engaged in which activities. That is, we want you to completely fill in all of the truth values in the twenty-element table for the binary relation E. Note that the table should be filled in so that the item in row $d \in D$ and column $a \in A$ should be the value of $E(d, a)$

We have already filled in the 4 values that were calculated in Question 2.

$D \setminus A$	DW	HCS	RB	WP
b	1 [3]	1 [3]	**0** [1]	**0** [1]
c	0 [4]	0 [7]	1 [2]	1 [2]
l	0 [4]	0 [7]	**1** [1]	0 [6]
p	**0** [1]	1 [7]	1 [2]	1 [2]
r	1 [5]	1 [7]	1 [2]	1 [2]

Using quantifier rules on Statements I-VI, justify each of the twenty answers in your table. (We have entered the four answers give to you in Question 2.)

The best way to present your solutions is by giving a text narrative (using either English or symbols) that includes any intermediate results you need on the way to get your answers.

Then you may indicate where each of the answers occur in your narrative.

Your text narrative should start every new argument on a new line with double spacing between the conclusion of one argument and the start of the next.

Hint: It may (or may not) help you to determine *how many* activities each of the dogs engaged in.

qr1. $\neg E(p, DW) \wedge E(l, RB) \wedge \neg E(b, RB) \wedge E(b, WP)$ given by Q2

qr2. by Statement III, all golden retrievers (c, P, r) engaged in RB and WP

qr3. by Statement VI, blaze engaged in exactly 2 activities, and since they didn't engage in RB or WP, the only 2 other activities are DW & HCS, so $E(b, DW) \wedge E(b, HCS)$

qr4. by Statement IV, since Phoebe engaged in every activity that Clover did, if Clover engaged in DW, then Phoebe would have had to engaged in DW, which contradicts qr1, therefore Clover couldn't have engaged in DW. by Statement IV, since Clover engaged in every activity Lulu did, if Lulu engaged in DW, then Clover would have had to engage in DW, which contradicts what we proved earlier that Clover couldn't have engaged in DW, therefor Lulu couldn't have engaged in DW.

Family Name: H

q15. Since no 2 dogs engaged in all of the same activities (by Statement II), the 3 goldens (C, P, R) who have all done RB & WP must all have engaged in the combination of DW & HCS Differently. There are 4 different cases: engage in neither, engage in both, or engage in 1 but not the other:

DW	HCS	
0	0	since C & P both didn't do DW, they must be one of these two, which means R must be one of these two, both of which include engaged with DW; therefore E(, DW)
0	1	
1	0	
1	1	

q16. by Statement II, no two dogs engaged in all of the same activities; if Lulu engaged in WP then 3 dogs would all have engaged in DW, RB, WP, but with only 1 activity left and only two states, that causes a contradiction, therefore ~E(, WP)

q17. by Statement IV(4) by transitivity contradiction

Family Name: Hansen

Question 4 (20): (Binary Relations on a Set) Parts (a) and (b) deal with two binary relations P and Q, each from a set to itself. Part (a) is on this page, part (b) on the next.

(a, 10) Let $A = \{a, b, c, d\}$. The relation $P \subseteq A \times A$ is a **partial order**. ⇒ transitive, antisymmetric, Reflexive
It is known that $(a, b), (b, c), (d, c) \in P$, and $(a, d) \notin P$.

(i) In the table given below, label all pairs that must be in P with a "✓" and all pairs that cannot be in P with a "×". For pairs that are neither, leave their entry blank. We start you off by labeling the four known pairs with the appropriate "✓" or "×".

Justify your answers. That is, you must provide a brief explanation as to why each of the pairs you marked "✓" are in P and those you marked "×" are not in P. Each pair that you justify must be explicitly named as being justified.

Your explanations must reference the properties of P that are being used. They must also be consistent. That is, if you somewhere use the fact that $(x, y) \notin P$ where $(x, y) \neq (a, d)$, then you must have already previously proven that $(x, y) \notin P$.

(ii) How many partial orders P' exist that satisfy $(a, b) \in P'$, $(b, c) \in P'$, $(d, c) \in P'$, and $(a, d) \notin P'$? Briefly explain how you know this. Draw a Hasse diagram illustrating each such P' that could exist.

(a,a)	(a,b)	(a,c)	(a,d)	(b,a)	(b,b)	(b,c)	(b,d)	(c,a)	(c,b)	(c,c)	(c,d)	(d,a)	(d,b)	(d,c)	(d,d)
✓ 2	✓ 1	✓ 4	× 1	× 3	✓ 2	✓ 1	× 5	× 4	× 3	✓ 2	× 3			✓ 1	✓ 2

1. Given $P(a,b) \wedge P(b,c) \wedge P(d,c) \wedge \neg P(a,d)$

2. By Reflexivity (partial order) all identity pairs must exist.
$P(a,a) \wedge P(b,b) \wedge P(c,c)$

3. By antisymmetry (partial order), the reverse pair of any existing pair does not exist (except identity pairs). $\neg P(b,a) \wedge \neg P(c,b) \wedge \neg P(c,d)$

5. By transitivity (partial order), the unconfirmed existence of some pairs would cause contradictions with confirmed existing pairs, and therefore cannot exist. $(P(a,b) \wedge \neg P(a,d)) \rightarrow \neg P(b,d)$

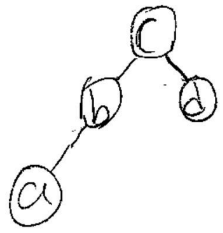
4. By transitivity (partial order): $(P(a,b) \wedge P(b,c)) \rightarrow P(a,c)$
and by antisymmetry (partial order): $P(a,c) \rightarrow \neg P(c,a)$

ii)

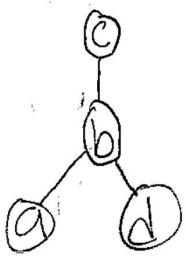
There are exactly [3] partial orders that satisfy the conditions given. The original is (1) (all of the values proven in Part i with the unknowns set to false; making P(d,b) true has no effect on the other possible values, so adding P(d,b) makes (2); you can't add P(d,a) without also including P(d,b) through transitivity, so both set to True makes (3).

Hasse diagrams

all false (original)



just P(d,b)



P(d,a) which forces P(d,b) through transitivity



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(b, 10) Let $B = \{a, b, c, d, e\}$. The relation $Q \subseteq B \times B$ is an **equivalence relation**. It is known that $(a, c), (b, c) \in Q$, and $(c, d), (c, e) \notin Q$. symmetry, Reflexive, transitive

(i) In the table given below, label all pairs that must be in Q with a "✓" and all pairs that cannot be in Q with a "×". For pairs that are neither, leave their entry blank. We start you off by labeling the four known pairs with the appropriate "✓" or "×".

Justify your answers following the same rules that were given in part (a).

(ii) How many Equivalence Relations Q' exist that satisfy $(a, c) \in Q'$, $(b, c) \in Q'$, $(c, d) \notin Q'$, $(c, e) \notin Q'$? Briefly explain how you know this. Write down the possible equivalence relation(s) in partition form, that is, as a set of sets of items in B.

(a,a)	(a,b)	(a,c)	(a,d)	(a,e)		(b,a)	(b,b)	(b,c)	(b,d)	(b,e)
✓ 2	✓ 4	✓ 1	× 5	× 5		✓ 4	✓ 2	✓ 1	× 5	× 5
(c,a)	(c,b)	(c,c)	(c,d)	(c,e)		(d,a)	(d,b)	(d,c)	(d,d)	(d,e)
✓ 3	✓ 3	✓ 2	× 1	× 1		× 5	× 5	× 3	✓ 2	
(e,a)	(e,b)	(e,c)	(e,d)	(e,e)						
× 5	× 5	× 3		✓ 2						

i)

1. given $P(a,c) \wedge P(b,c) \wedge \neg P(c,d) \wedge \neg P(c,e)$

2. by Reflexivity (eq. Rel.): $P(a,a) \wedge P(b,b) \wedge P(c,c) \wedge P(d,d) \wedge P(e,e)$

3. by Symmetry (eq. Rel.): $(P(a,c) \rightarrow P(c,a)) \wedge (P(b,c) \rightarrow P(c,b)) \wedge (\neg P(c,d) \wedge \neg P(c,e)) \rightarrow (\neg P(d,c) \wedge \neg P(e,c))$

4. by transitivity (eq. Rel.) $\{(P(a,c) \wedge P(c,b)) \rightarrow P(a,b)) \wedge ((P(b,c) \wedge P(c,a)) \rightarrow P(b,a))\}$

5. by transitivity (eq. rel.): $(\neg P(c,e) \wedge \neg P(c,d) \wedge P(a,c) \wedge P(b,c)) \rightarrow \{\neg P(a,d), \neg P(a,e), \neg P(b,d), \neg P(b,e)\}$
be symmetry (eq. rel): $(\neg P(a,d) \rightarrow \neg P(d,a)) \wedge (\neg P(a,e) \rightarrow \neg P(e,a)) \wedge (\neg P(b,d) \rightarrow \neg P(d,b)) \wedge (\neg P(b,e) \rightarrow \neg P(e,b))$

ii) there are only two possible eq. rel.s Q' that satisfy the given conditions: either d & e are equal, or their not (not both)

1. $\{(a,b,c)\}, \{(d)\}, \{(e)\}$

2. $\{(a,b,c)\}, \{(d),(e)\}$

Family Name: Hansen

Family Name: Hansen

Question 5 (20+5): (Number Theory)

- (a, 10) The naturals 96 and 43 are relatively prime.
  Use the Extended GCD algorithm as taught in class to find integers a and b satisfying the equation $a \cdot 96 + b \cdot 43 = 1$. Show all of your work. You must use the same formatting structure as shown in class.

  After solving the problem, write your final solution here
  $a =$ 8. $b =$ -14

$96 \% 43 = 10$ | $10 = 1(96) - 2(43)$
$43 \% 10 = 3$ | $3 = 2(96) - 4(43$
$10 \% 3 = 1$ |

- (b, 5) Using the results from the previous part, determine both an inverse of 96 modulo 43 and an inverse of 43 modulo 96.
  For full credit, your answers should each be the smallest natural numbers that are inverses.

  After solving the problem, write your final solutions here:
  129 is an inverse of 96 modulo 43.
  86 is an inverse of 43 modulo 96.

- (c, 5) You are now given that $6 \cdot 50 - 13 \cdot 23 = 1$.
  Using the technique taught in class, determine the smallest natural x that solves both the congruences

  $$x \equiv 3 \pmod{50} \quad \text{and} \quad x \equiv 4 \pmod{23}.$$

  Show all of your your work.
  After solving the problem, write your final solution below:

  The smallest natural satisfying both congruences is $x =$ 1.

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- (d, 5 extra credit) Find all naturals x in $\{1, 2, 3, \ldots, 34\}$ that satisfy $x^2 \equiv 1 \pmod{35}$.
  Hint: You may find the following facts useful:
  1. $x^2 - 1 = (x+1)(x-1)$.
  2. For any prime number p and naturals a, b: p divides ab if and only if p divides a or p divides b.
  3. $3 \cdot 7 - 4 \cdot 5 = 1$.

  Show all of your your work. To receive full credit, you need to prove that they are *all* the possible values.
  After solving the problem, write your final solution below:

  All possible values of x are 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17, 18, 19, 22, 23, 24, 26, 27, 29, 31, 32, 33, 34

$1^2 = 1 \neq$

$2^2 = 4 \neq$

$3^2 = 9 \neq$

$4^2 = 16 \neq$

35

$7 \cdot 5 \cdot 35$

x^2 multiples of 5, 7 are out

Family Name: Hansen

Question 6 (20): The following are ten true/false questions, with no explanation needed or wanted, no partial credit for wrong answers, and no penalty for guessing.

After reading the questions, write the correct answer, either T (for true) or F (for false), in the corresponding column.

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
T	F	F	T	F	T	T	T	T	F

(a) It is always the case that if $\forall x : \forall y : R(x, y)$ is true, then $\exists x : \exists y : R(x, y)$ is true.

(b) The relation "$<$" on the naturals is anti-symmetric, i.e., if $x < y$ and $y < x$, then $x = y$.

(c) If a relation is reflexive, not symmetric, and transitive, then it must be a partial order.

(d) Let f be a function from X to Y, and g from Y to Z, be two injections (one-to-one functions). Then their composition $g \circ f$ must also be an injection.

(e) Let u and v be two non-empty strings over the alphabet $\{a, b\}$. If u is both a prefix of v and a suffix of v, then u and v must be the same string, that is, $u = v$.

(f) For all naturals x, y, and z, $x\%(y\%z) = (x\%y)\%z$, where "%" is the modular division operation, as in Java or Python.

(g) Let A be a finite set. Then the empty string λ is a member of the language A^* if and only if A is non-empty.

(h) The statement $X \Delta Y = (X \cup Y) \setminus (X \cap Y)$ is a set identity.

(i) Let X be the set of all even naturals, and let Y be the set of all odd naturals. Then there exists a set Z such that $Z \subseteq X$ and $Z \subseteq Y$.

(j) The following is a valid proof to show that no natural can be both even and odd. Let x be a number that is both even and odd. By the definitions, we know that $\exists y : x = 2y$ and $\exists y : x = 2y+1$. Instantiating both of these expressions to y as a, we get $x = 2a = 2a+1$ from which we derive $0 = 1$, a contradiction.